Name: ....

Reg. No: ....

APJ ABDULKALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M.TECH. DEGREE EXAMINATION JANUARY 2025

Course Code: 221ECS004

Course Name: Computational Intelligence

Max. Marks: 60 **Duration: 2.5 hours**

Branch: Computer Science and Engineering

PART A

(Answer all questions in PART A. Each question carries 5 marks)

Q. No.		Marks
1	Explain the Max-Min method and Max-Product method of fuzzy composition.	(5)
2	What is meant by the term defuzzification? Briefly describe the most common method of defuzzification?	(5)
3	What is the significance of mutation in a genetic algorithm?	(5)
4	Briefly explain the basic foraging behaviour of real ants that is modelled to create the artificial ant colony systems for solving optimization problems.	(5)
5	Briefly describe the basic Particle Swarm Optimization (PSO) algorithm.	(5)

PART B

(Answer any five full questions in PART B. Each full question carries 7 marks)

6 a. Let U = Flowers = {Jasmine, Rose, Lotus, Daffodil, Sunflower, Hibiscus, Chrysanthemum} be a universe on which two fuzzy sets, one of Beautiful flowers and the other one of Fragrant flowers are defined as shown below. (3)

P = Beautiful flowers = {(Jasmine, 0.3), (Rose, 0.9), (Lotus, 1.0), (Daffodil, 0.7), (Sunflower, 0.5), (Hibiscus, 0.4), (Chrysanthemum, 0.6)}

Q = Fragrant flowers = {(Jasmine, 1.0), (Rose, 1.0), (Lotus, 0.5), (Daffodil, 0.2), (Sunflower, 0.2), (Hibiscus, 0.1), (Chrysanthemum, 0.4)}

Compute the fuzzy sets $P \cup Q$, $P \cap Q$, $P - Q$.

b. Consider two fuzzy sets P and Q given by: (4)

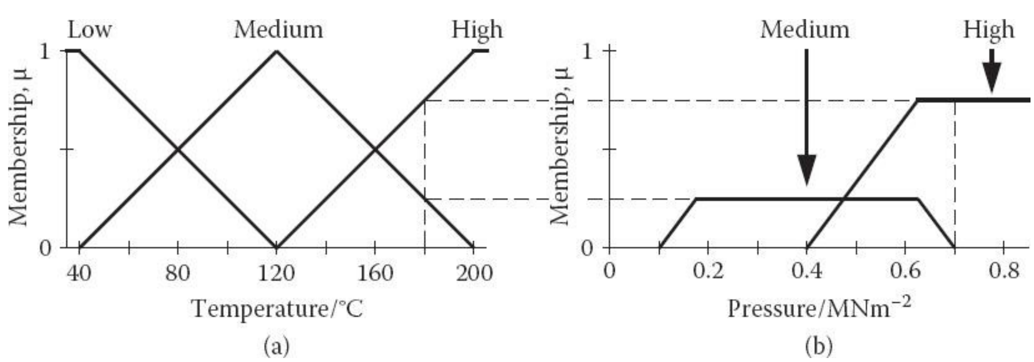
$$A = \left\{\frac{0.5}{Winter}, \frac{0.33}{Spring}, \frac{0.52}{Summer}, \frac{0.25}{Fall}\right\}$$

$$B = \left\{\frac{0.10}{Winter}, \frac{0.55}{Spring}, \frac{0.90}{Summer}, \frac{0.20}{Fall}\right\}$$

$$C = \left\{\frac{0.22}{High}, \frac{0.55}{Medium}, \frac{0.44}{Low}\right\}$$

Derive the following relations:
(a) If x is A or y is B then z is C.
(b) If x is A and y is ~B then z is C.

7 Consider a rule base that contains the following fuzzy rules: if temperature is high then pressure becomes high. if temperature is medium then pressure becomes medium. if temperature is low then pressure becomes low. Membership functions for pressure derived from these rules for a temperature of 180°C is depicted below. (7)



How will you proceed from here to obtain a defuzzified value as output.

8 Consider the following hyperparameter settings for a Genetic Algorithm implementation for solving the Travelling Salesman Problem: (7)

NUMCITIES = 20

EliteNumber = 3

ToursPerGeneration = 150

NumberGenerations = 10000

TournamentNumber = 5

ProbCrossover = 0.9

ProbMutation = 0.2

State the implications of each of these parameters on the search process for finding an acceptable solution.

9 Consider a particle swarm optimization system composed of three particles and a maximum velocity of 12. Assume that both random numbers r1 and r2, used for computing the movement of the particle towards the individual best position and social best position, are 0.6. Also, assume that the solution space is two-dimensional and real-valued. The current state of the swarm is as follows: (7)

Position of particles: x1 = (5,5); x2 = (9,2); x3 = (7,8)

Individual best positions: x1 = (5,5); x2 = (8,2); x3=(6,7)

Velocities: v1 = (3,3); v2 = (2,2); v3 = (5,5)

What will be the next position of each particle after one iteration of the PSO algorithm if the inertia parameter ω used in the velocity update formula is 0.7?

10 Describe Ant Colony System. What are the different types of Ant systems? (7)

11 Describe the commonly used fuzzy membership functions. (7)

12 Explain the concept of velocity clamping in Particle Swarm Optimization (PSO). How does velocity clamping improve the rate of convergence, and what impact does it have on the optimization process? (7)